



Verified Carbon Standard

NAM HOUNG - 1 HYDRO POWER PROJECT



Document Prepared By: 4K Earth Science Private Limited

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Summary:

4K Earth Science Private Limited (4KES) has been contracted by “Kosher Climate India Private Limited” to undertake verification and certification for the greenhouse gas (GHG) emission reductions reported from “Nam Houg – 1 Hydro Power Project” for the monitoring period 01-September-2023 to 31-December-2024 (Inclusive of both the dates) in the initial monitoring report version 1.0, date 06/04/2025.

The purpose of this project activity is to generate electricity through renewable Hydro energy source. The net electricity generated from this project is supplied to the national grid via the transmission line.

During this monitoring period, the project has operated continuously and satisfactorily. No changes on the project entity or applicable legislation have occurred during this fifth monitoring period.

Verification team has verified the technical details of the project equipment's/capacity of the project activity during Onsite audit, technical specification /10/ and photographs /14/ etc.

The project activity generated renewable electricity without any GHG emissions, which displaced part of the electricity otherwise supplied by grid. Thus, GHG emission reductions can be achieved.

The project activity is a hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houg - 1 Hydro Power Co., Ltd. Project was commissioned on 01-September-2023. Implementation status of the project activity has been verified during the On-site audit by checking the supportive evidences of Commercial operation certificate/19/

The project is expected to constantly contribute clean energy to the Lao Power Grid. It not only conforms to the present situation, but also can adapt to future changes in the electricity market. The baseline scenario of the project is continuation of the present situation, i.e. electricity supplied from the power grid.

The site is located in the Xayabouly Province, Lao PDR as confirmed during the on-site audit. The project geo-coordinates are: 19°11'19.7"N; 101°48'17.8"E as provided under section 1.8 of the VCS MR /1/. Verification team has verified the physical address through internet /07/ by enrolling the coordinates and found to be consistent.

During current monitoring period i.e. from 01-September-2023 to 31-December-2024, the project activity has supplied 72,978 MWh net electricity to the grid, there by resulted in emission reduction of 26,298 tCO_{2e}. The monitoring period subject to this monitoring report is inclusive of first and last day of period.

A risk-based approach has been followed to perform the First periodic verification of the project activity. In the course of the verification, 04 Corrective Action Requests (CARs) and 00 Clarification (CL) were raised.

The management of the “Kosher Climate India Private Limited” responsible for the preparation of the GHG emissions data and the reported GHG emissions reductions on the basis set out within the project final monitoring report Version 2 dated 14/05/2025. The calculation and determination of GHG emission reductions from the project is the responsibility of the management of the Kosher Climate India Private Limited. The development and maintenance of records and reporting procedures are in accordance with the Monitoring Report Version 2 dated 14/05/2025.

It is our responsibility to express an independent GHG verification opinion on the GHG emissions and on the calculation of GHG emission reductions from the project for the period 01-September-2023 to 31-December-2024 based on the reported emission reductions in the final monitoring report Version 2 dated 14/05/2025 for the same period.

Based on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these, 4KES planned and performed our work to obtain the information and explanations that we considered necessary to provide sufficient evidence for us to give reasonable assurance that this reported amount of GHG emission reductions for the period is fairly stated.

4KES confirms the following;

Reporting period: 01-September-2023 to 31-December-2024

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
01-Sep-2023 to 31-Dec-2023	10,857	2,170	0	8,687	0	8,687
01-Jan-2024 to 31-Dec-2024	22,010	4,398	0	17,611	0	17,611
Total	32,867	6,568	0	26,298	0	26,298

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1 INTRODUCTION

1.1 Objective

4KES has been commissioned by 'Kosher Climate India Private Limited' to perform verification of its registered VCS project 'Nam Houg – 1 Hydro Power Project', for the reported GHG emission reductions for the given monitoring period 01-September-2023 to 31-December-2024 (both dates included). The VCS projects must undergo an independent third-party verification and certification of emission reductions as the basis for issuance of Voluntary Emission Reductions (VERs).

The objectives of this verification exercise are, by review of objective evidence, to establish that:

- The project activity has been implemented and operated as per the project description (PD) and that all physical features (technology, project equipment and monitoring and metering equipment) of the project are in place;
- Monitoring report and other supporting documents are complete;
- The data is recorded and stored as per the monitoring methodology and approved monitoring plan.
- To confirm that the monitoring system is implemented and fully functional to generate Voluntary Emission Reductions (VERs/VCUs) without any double counting, and
- To establish that the data reported are accurate, complete, consistent, transparent and free of material error or omission by checking the monitoring records and the emissions reduction calculation.

1.2 Scope and Criteria

The scope of verification is to assess the claims and assumptions made in the VCS monitoring report (MR) against the VCS criteria, including but not limited to, VCS standard/3/, applied methodology and other relevant rules and requirements established for VCS project activities.

The Verification is not meant to provide any consulting towards the project participants. However, stated requests for clarification and/or correction actions request may have provided inputs for improvement of the project design.

1.3 Level of Assurance

The verification team verified the complete monitoring data /11/ and /12/ for the monitoring parameter of the monitoring plan against the Monitoring report/1/ and ER Calculation sheet /2/ and confirms that the reported emission reductions are free from any type of material errors. On-site audit/Physical site visit were also conducted to verify the implementation and monitoring plan of the project activity. The implementation of the project activity (i.e. major equipment and

the metering arrangement) was verified during the On-site audit/physical site visit and found in accordance with the monitoring report /1/ and registered PD /05/. Verification team has also verified the technical details of the project equipment's and metering arrangement with the supportive documents /08/, /10/, and /14/ and found correct. Therefore, 4KES confirms that the verification is conducted with reasonable level of assurance.

1.4 Summary Description of the Project

The purpose of this project activity is to generate electricity through renewable Hydro energy source. The net electricity generated from this project is supplied to the national grid via the transmission line.

During this monitoring period, the project has operated continuously and satisfactorily. No changes on the project entity or applicable legislation have occurred during this first monitoring period.

Verification team has verified the technical details of the project equipment's/capacity of the project activity by On-site audit, technical specification /10/ and photographs /14/ etc.

The project activity generated renewable electricity without any GHG emissions, which displaced part of the electricity otherwise supplied by grid. Thus, GHG emission reductions can be achieved.

The project activity is a hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houg - 1 Hydro Power Co., Ltd. Project was commissioned on 01-September-2023. Implementation status of the project activity has been verified during the On-site audit by checking the supportive evidences of Commercial operation certificate/19/

The project is expected to constantly contribute clean energy to the Lao Power Grid. It not only conforms to the present situation, but also can adapt to future changes in the electricity market. The baseline scenario of the project is continuation of the present situation, i.e. electricity supplied from the power grid.

The site is located in the Xayabouly Province, Lao PDR as confirmed during the on-site audit. The project geo-coordinates are: 19° 11' 19.7"N; 101° 48' 17.8"E as provided under section 1.8 of the VCS MR /1/. Verification team has verified the physical address through internet /07/ by enrolling the coordinates and found to be consistent.

During current monitoring period i.e. from 01-September-2023 to 31-December-2024, the project activity has supplied 72,978 MWh net electricity to the grid, there by resulted in emission reduction of 26,298 tCO₂e. The monitoring period subject to this monitoring report is inclusive of first and last day of period.

2 VERIFICATION PROCESS

The registered VCS project is undergoing 1st verification under VCS (1st Crediting period), the approach adopted to ensure the quality of emission reductions is described in the following sections.

2.1 Method and Criteria

4KES assessed and determined whether the proposed implementation and operation of the project activity and the steps taken to report emission reductions comply with the criteria and relevant guidance provided by the VCS Board. The validation/verification process consist of the following three phases;

- A desk review of the VCS PD and VCS MR
- follow up interviews with project stakeholders
- The resolution of outstanding issues and issuance of final report and opinion.

The prepared verification report and other supporting documents then undergo an internal quality control before being submitted to the VCS executive board for issuance of credits as per VCS standard v4.7/3/ and VCS program guide/4/.

Duration of Verification:

Verification Contract	25/03/2025
Site Visit Date	21/04/2025
Draft Verification Report	05/05/2025
Final Verification Report	20/05/2025

2.2 Document Review

The verification is performed primarily as a document review of the approved PD and associated documents as stated in detail in appendix II of this document. The assessment is performed by a verification team using a protocol. The cross checks between information provided in the Monitoring report, PD and information from sources other than those used, if available, the team's sectoral or local expertise and, if necessary, independent background investigations.

2.3 Interviews

Verification team performed the On-site audit and interviewed PP representatives/Employees/O&M Team/Stakeholders and reviewed documents to achieve a reasonable level of assurance in the verification. No sampling procedures were adopted in document verification and all the document were cross checked to ensure conservative estimation of emission reduction. Kindly find below names of the persons interviewed during Onsite audit.

Location	The site is located in Xayabouly Province, Lao PDR The project geo-coordinates are: 19° 11'19.7"N; 101° 48'17.8"E.		
Dates	21/04/2025 (Onsite audit by Team Leader and Technical Expert)		
Key points discussed	Name of person interviewed	Designation, Organization	
Project Implementation,	Sorlaphon	Deputy Director, Nam Houg - 1 Hydro Power Co., Ltd	

Operational data, Calibration, Data collection, QA/QC, procedures, Calculation of ERs, VCS requirements.	Savad Douangdara	Manager, Nam Houng - 1 Hydro Power Co., Ltd
	Hong nai ban	Local people
	kam ma kan	Local people
	Hong Nuay	Local people
	Naveen Kumar	Kosher Climate India Private Limited
Sustainable development Impact, Complaints, Mode of communication, Environmental impact any other concerns		

2.4 Site Visits

Verification team performed the On-site audit and interviewed PP representatives/Employees/O&M Team/Stakeholders and reviewed documents to achieve a reasonable level of assurance in the verification. No sampling procedures were adopted in document verification and all the document were cross checked to ensure conservative estimation of emission reduction. The name of the persons interviewed is reported under section 2.3 of this report.

2.5 Resolution of Findings

The objective of this step is to identify, discuss and conclude on the issues related to the monitoring, implementation and operations of the registered project activity that could impair the capacity of the registered project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions. This is done based on the desk review and on-site audit/Physical site visit. The verification team prepares and/or updates a verification protocol (internal document) that records the conformities and non-conformities, which may be of following types;

CAR (Corrective Action Request) is raised if one of the following occurs:

- Non-compliance with the monitoring plan, the methodology or the standardized baseline are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;

- Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project participants;
- Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;
- Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

Clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met. All CARs and CLs raised by the 4KES during verification shall be resolved prior to submitting a request for issuance.

FAR (Forward Action Request) is raised during verification if the monitoring and reporting require attention and/or adjustment for the next verification period.

During the Verification process, total 04 CAR and 00 CL were raised and resolved satisfactorily. No FAR has been raised in the verification. The list of CARs/CLs/FARs raised and the response provided, the mean of validation, reasons for their closure and references to correction in the relevant documents are provided in Appendix III of this report.

2.5.1 Forward Action Requests

No FAR raised during the verification.

2.6 Eligibility for Validation Activities

4KES conducted the verification activity; the validation was performed by the other VVB. 4KES has a valid UNFCCC accreditation in the sectoral scope from UNFCCC. The accreditation scope can be checked from the below link: <http://cdm.unfccc.int/DOE/list/DOE.html?entityCode=E-0069>.

3 VALIDATION FINDINGS

3.1 Methodology Deviations

No deviation is applied during the monitoring period or previous monitoring period.

3.2 Project Description Deviations

There is no change in project design related to installed capacity, technology, specification, etc.

3.3 New Project Activity Instances in Grouped Projects

The project is not a grouped project. Therefore, this section is not applicable.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes ☒ No

4 VERIFICATION FINDINGS

4.1 Project Details

The purpose of this project activity is to generate electricity through renewable Hydro energy source. The net electricity generated from this project is supplied to the national grid via the transmission line.

During this monitoring period, the project has operated continuously and satisfactorily. No changes on the project entity or applicable legislation have occurred during this fifth monitoring period.

Verification team has verified the technical details of the project equipment's/capacity of the project activity by On-site audit, technical specification /10/ and photographs /14/ etc.

The project activity generated renewable electricity without any GHG emissions, which displaced part of the electricity otherwise supplied by grid. Thus, GHG emission reductions can be achieved.

The project activity is a hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houng - 1 Hydro Power Co., Ltd. Project was commissioned on 01-September-2023. Implementation status of the project activity has been verified during the On-site audit by checking the supportive evidences of Commercial operation certificate/19/

The project is expected to constantly contribute clean energy to the Lao Power Grid. It not only conforms to the present situation, but also can adapt to future changes in the electricity market. The baseline scenario of the project is continuation of the present situation, i.e. electricity supplied from the power grid.

The site is located in the Xayabouly Province, Lao PDR as confirmed during the on-site audit. The project geo-coordinates are: 19° 11'19.7"N; 101° 48'17.8"E as provided under section 1.8 of the VCS MR /1/. Verification team has verified the physical address through internet /07/ by enrolling the coordinates and found to be consistent.

During current monitoring period i.e. from 01-September-2023 to 31-December-2024, the project activity has supplied 72,978 MWh net electricity to the grid, there by resulted in emission reduction of 26,298 tCO₂e. The monitoring period subject to this monitoring report is inclusive of first and last day of period.

During the process of verification, 4KES confirmed the capacity, power generation, arrangement for evacuation of electricity generated, technical specifications, date of commissioning and necessary clearances for setting the project activity. List of documents reviewed during the course of verification is presented under Appendix II of this report.

The selected monitoring period i.e. 01-September-2023 to 31-December-2024 (both days included) is within the 1st crediting period i.e. 01-September-2023 to 31-August-2030 which is accepted to the verification team.

There was no major breakdown or shutdowns during the monitoring period which might affect the applicability of methodology or might cause material errors in emission reductions as confirmed from the interviews and monthly electricity records /11/.

The assessment team confirmed that there is no proposed or actual change to the project design during this monitoring period. The project design as mentioned in the registered PD & monitoring report submitted is implemented and thus the same is acceptable to the assessment team. All required monitoring equipment's and procedures as mentioned in the registered PD & monitoring report are available and implemented in an appropriate manner.

The organisational role and responsibility as mentioned in the registered PD /05/ & monitoring report /1/ is followed onsite as confirmed from the interviews and organization structure /09/. All the emergency preparedness as mentioned in the registered PD & monitoring report is followed onsite and no discrepancies were found regarding the same.

The assessment team found that the project is in line with the registered PD, monitoring report, and no deviation on project design or monitoring plan is observed.

As per the registered PD /05/, ex-ante emission reductions for this monitoring period are 42052tCO₂e however actual emission reductions corresponding to this monitoring period are 26,298 tCO₂e. The actual emission reduction is lower than the estimated emission reduction and hence accepted to the verification team.

Section 1.2 of the VCS MR /1/ mention about the audit history. The registered VCS project is undergoing 1st periodic verification for the monitoring period 01-September-2023 to 31-December-2024 (including both the days) under VCS 1st Crediting period i.e. 01-September-2023 to 31-August-2030 (including both the days). Verification team has checked the audit history table mentioned under section 1.2 of the VCS MR /1/ and found accurate.

Item	Evidence gathering activities, evidence checked, and assessment conclusion:
Audit history	Verification team has checked the audit history mentioned under section 1.2 of the MR/1/ and found that MR/1/ has been filled appropriately.
Double counting and participation under other GHG programs	Verification team confirms that it has checked that the project activity is neither registered nor applied for registration in other GHG or Non GHG program(s) to obtain the certified GHG emission reductions. The Project Developer is here by declaring that the emission reductions claimed under this project activity will not be double counted under any other GHG program. The declaration/15/ on the same has been provided and found appropriate. This was confirmed by the validation team by verifying the declaration provided by PP and checked with different registry also. Thus, emission reductions generated by project will be solely claimed by PP and PP has the right of use, which is acceptable.

No double claiming with emissions trading programs or binding emission limits	The project reductions and removals or project activities is not included in an emissions trading program or binding emission limit. The same is confirmed by the VVB by checking other registries like GS, CDM and GCC.
No double claiming with other forms of environmental credit	The project activity not sought, received, or is planning to receive credit from another GHG-related environmental credit system which is confirmed by the PP during the On-site audit interview and PP has submitted the declaration/15/.
Supply chain (scope 3) emissions double claiming	Project proponent or authorized representative is not a buyer or seller of a product whose emissions footprint is changed by the project activities. The same is confirmed from the PP's representative interviews.
Sustainable development contributions	Verification team has checked the supportive for the activities that result in the SD contributions from the project activity for each SDG. SDG Target 7.2: During the monitoring period, a total of 72,978MWh of renewable energy was delivered to grid. Verification team has verified the reported electricity export/import data with the monthly records/11/ and found consistent. The electricity supplied to grid is also cross-checked from the monthly invoices/12/ and found consistent. SDG Target 8.5: The project activity provided 17 new employment opportunities. Verification team has verified the same during the interviews and list of employees/16/. SDG Target 13.0: During the monitoring period, project has avoided the emission of 26,298 tCO₂e. Verification team has checked the calculation from the monitoring report/1/ and the ER calculation sheet/2/ and found correct. Project activity provides local jobs. Section 1.12 of the Monitoring report has been checked and found OK. The project activity contributes positively to sustainable development which has been verified from the interviews of local people (Section 2.3 of this report) and supportive documents as mentioned above.
Additional information relevant to the project	No commercially sensitive information has been excluded from the public version of the monitoring report.

Opinion:

Assessment team concludes the following:

- a) There are no material discrepancies between project implementation and the project description provided in the PD /05/.
- b) The monitoring plan is implemented completely and monitoring system (i.e., process and schedule for obtaining, recording, compiling and analysing the monitored data and parameters) is appropriate.
- c) There are no material discrepancies between the actual monitoring system and the monitoring plan set out in the project description and the applied methodology.
- d) The GHG emission reductions or removals generated by the project have not included in an emissions trading program or any other mechanism that includes GHG allowance trading.
- e) The project has not received or sought any other form of environmental credit, or has become eligible to do so since validation or previous verification.
- f) The project is registered under VCS only. No issuance is sought for this monitoring period under CDM as confirmed from the CDM webpage (<https://cdm.unfccc.int/Projects>) and declaration given by the PP /15/. Hence Verification team confirms that there is no double counting associated with project activity being participation of other GHG programs.

In view of the information's as verified above the assessment team is able to conclude that the project has been implemented as described in the project description.

4.2 Safeguards and Stakeholder Engagement

4.2.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	All stakeholders were identified at validation. During the On-site audit, the assessment team interviewed PP's representative/stakeholders and confirmed that no local stakeholder consultation was conducted for this project activity for this monitoring period. Thus, this section is not applicable.
Legal or customary tenure/access rights	This section is not applicable. The project is built and owned by Nam Houng – 1 Hydro Power Co., Ltd. Nam Houng - 1 Hydro Power Co., Ltd. is a SPV of Simoung Group Co., Ltd. Simoung Group Co., Ltd. is the legal owner of the Nam Houng – 1 Hydro Power Project, operated

	under Nam Houng - 1 Hydro Power Co., Ltd. The company holds the legal rights to control and manage the project's activities. The business license and related documentation serve as evidence of these rights. During the verification process, the assessment team reviewed the Commissioning certificate/19/ and confirmed its validity and alignment with the stated ownership and operational claims.
Stakeholder diversity and changes over time	Stakeholders were identified at validation, including local authorities, local people with social, economic, and cultural similarities. The project anticipates no significant changes to the group's cultural integrity in the project region. Stakeholder groups have not changed over time. The same is confirmed during the On-site audit/site visit observation and local stakeholder interviews.
Expected changes in well-being	The project contributes to Environmental, Economic and Social well-being by upgraded the existing roads, directly contributing to the improvement of infrastructure in region; created jobs with good working conditions for local people during the operation phase. The same is confirmed during the On-site audit/site visit observation, employee interviews and list of employees /16/.
Location of stakeholders	The verification team confirms that all community members—both those directly and indirectly impacted by the project—are primarily local residents living near the project areas. This was verified through interviews conducted with local people during the onsite assessment of the project.
Location of resources	The verification team confirms that location of territories and resources to which stakeholders have customary access is the nearby land in Xayabouly Province, Lao PDR.

4.2.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Ongoing consultation	PP has a mechanism for on-going communication with local stakeholders which has been discussed during the on-site audit. During the monitoring period (from 01-September-2023 to 31-December-2024), there are no comments received for the project

	activity which was confirmed during the interview of PP representative & Local stakeholders, comment book /17/. Mechanism for on-going communication with local stakeholders include: - Direct communication at the PP office. People can go directly to the site to lodge their complaint. The person in charge of the power plant is available on site to take note of their complaint and record it in a dedicated notebook. Additionally, a grievance register has been set up to record comments from stakeholders - Electronic communication via the project owner's website or telephone. The same is confirmed from the On-site audit observation, local stakeholder interviews, comment book /17/ During current monitoring period, there are no grievances received related to this power plant. Thus, assessment team is of the opinion that the ongoing stakeholder mechanism is adequate and appropriate.
Date(s) of stakeholder consultation	21-August-2022
Communication of monitored results	During the onsite assessment, the verification team interviewed local peoples who confirmed that, during the LSC meeting held prior to the initial PD submission, the Project Proponent (PP) shared a dedicated email address, phone number, and physical address with all stakeholders to ensure open and accessible communication channels for future engagement. Additionally, a grievance register is maintained at the project site to record any stakeholder complaints.
Consultation records	A grievance register has been maintained at the site to document any complaints from stakeholders. The verification team reviewed the register and confirms that no complaints were recorded during this monitoring period.
Stakeholder input	Not Applicable, as no comments were received during this monitoring period.

4.2.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Consent	The free, prior and informed consent took place during the public consultations when the project was explained publicly. No ongoing or unresolved conflicts are going on. PP has a mechanism for on-going communication with local stakeholders which has been discussed during the On-site audit. During the monitoring period (from 01-September-2023 to 31-December-2024), there are no comments received for the project activity which was confirmed during the interview of PP representative & Local stakeholders, comment book /17/. During current monitoring period there are no grievances received related to this plant. Thus, assessment team is of the opinion that the ongoing stakeholder mechanism is adequate and appropriate.
Outcome of FPIC discussion	No ongoing or unresolved conflicts are going on which was confirmed during the interview of PP representative & Local stakeholders, comment book /17/.

4.2.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Grievance received and steps taken to resolve the grievance including the outcomes of the resolution	PP has a mechanism for on-going communication with local stakeholders which has been discussed during On-site audit. During the monitoring period (from 01-September-2023 to 31-December-2024), there are no comments received for the project activity which was confirmed during the interview of PP representative & Local stakeholders, comment book /17/. During current monitoring period there are no grievances received related to the plant. Thus, assessment team is of the opinion that the ongoing stakeholder mechanism is adequate and appropriate.
Grievance redress procedure	PP has a mechanism for on-going communication with local stakeholders which has been discussed during the On-site audit/physical site visit. During the monitoring period (from 01-

	September-2023 to 31-December-2024), there are no comments received for the project activity which was confirmed during the interview of PP representative & Local stakeholders, comment book /17/ During current monitoring period there are no grievances received related to this power plant. Thus, assessment team is of the opinion that the ongoing stakeholder mechanism is adequate and appropriate.
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4.2.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
During this monitoring period, no comment received for the project activity which was confirmed during the interview of PP representative & Local stakeholders, comment book /17/.	N/A	N/A

4.2.6 Risks to Local Stakeholders and the Environment

4.2.6.1 Management Experience

The project is managed by a highly experienced team with a strong track record in implementing similar projects and engaging local communities. The management team comprises experts in hydropower development, environmental management, and stakeholder engagement, with prior experience in successfully delivering sustainable energy projects. Their expertise includes compliance with international best practices, regulatory frameworks, and environmental and social safeguard policies. The Operation and Maintenance Department consisting of plant technicians undertakes the technical actions required by the monitoring plan, under the Project Director's authority, to collect and record related data. Verification team has found the PP justification acceptable.

4.2.6.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human induced	As PP mentioned in the monitoring Report under the section of 2.2.2, The project has positive impacts on the environment, society and economy;

risks to stakeholders' wellbeing	and has no risks to stakeholders' wellbeing. The same is confirmed from the On-site audit/site visit observation, local stakeholder interviews.
Risks to stakeholder participation	PP has a mechanism for on-going communication with local stakeholders which has been discussed during the On-site audit/physical site visit. During the monitoring period (from 01-September-2023 to 31-December-2024), there are no comments received for the project activity which was confirmed during the interview of PP representative & Local stakeholders, comment book /17/ During current monitoring period there are no grievances received related to the plant. Thus, assessment team is of the opinion that the ongoing stakeholder mechanism is adequate and appropriate.
Working conditions	There is no risk related to working conditions in the plant. This is also verified from on-site audit with PP's representative/employees, list of employees /16/, and training records /18/.
Safety of women and girls	There is no risk related to Safety of women and girls. This is also verified from on-site audit with PP's representative/employees, list of employees /16/.
Safety of minority and marginalized groups, including children	There is no risk related to Safety of minority and marginalized groups, including children. This is also verified from on-site audit with PP's representative/employees, list of employees /16/
Pollutants (air, noise, discharges to water, generation and release of hazardous materials and chemical pesticides and fertilizers	The project does not involve major construction activity. It primarily requires the installation of the Hydro power plant, transmission line, electricity meter and transformers etc. There is no risk due to generation of pollutants (noise, discharges to water, generation of waste, and release of hazardous materials), due to project activity. The project rather improves air quality by preventing fossil fuel burns and mitigates climate change by producing electricity from a clean source of energy, i.e. hydro energy. Overall, the project complies with environmental standards and poses no pollution-related threats to stakeholders or ecosystems.

4.2.7 Respect for Human Rights and Equity

4.2.7.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Discrimination	The PP strictly follows the national laws prohibiting discrimination and hence no discrimination has occurred or will occur in the future.
Sexual harassment	Project is not involved in discrimination and sexual harassment as confirmed during the On-site interviews with PP's representatives/employees, Grievance mechanism /17/.
Gender equity in labor and work	Women and men are both part of the team in charge of the plant and site management. There is no risk in gender equity and work as confirmed during the interviews with PP's representatives/employees and employment records /16/.
Forced labor	Project does not use victims of human trafficking, forced labour and child labour. The same is confirmed during the interviews with PP's representatives/employees.
Child labor	Project does not use victims of human trafficking, forced labour and child labour. The same is confirmed during the interviews with PP's representatives/employees and list of employees /16/.
Human trafficking	Project does not use victims of human trafficking, forced labour and child labour. The same is confirmed during the interviews with PP's representatives/employees.

4.2.7.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
Human rights	The project recognizes and respect the rights of workers and local stakeholders. The same is confirmed during the interviews with PP's representatives/employees.

4.2.7.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
Preservation and protection of cultural heritage	The project does not impact and protects any cultural heritage as part of project activities. No indigenous people are affected by the project. The same is confirmed during the interviews with PP's representatives/stakeholders.

4.2.7.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk identified	No property right risk applicable as the land / property for the project activity belongs to the project proponent and, it can be supported through the ownership documents. This ensures that their lands are property rights are protected and respected. The same is confirmed during the interviews with PP's representatives/stakeholders.

4.2.7.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Summary of the benefit sharing plan	N/A
Benefit sharing during the monitoring period	N/A

4.2.8 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	No risk identified as per the stakeholder interviews and this project activity doesn't result in adverse effects; instead, it positively impacts on the biodiversity and ecosystem by decreasing air pollution and mitigating climate change.
Soil degradation and soil erosion	As this project activity is Hydro power plant and doesn't result in soil degradation and soil erosion. Vegetative cover is maintained in exposed areas, while erosion control structures, such as silt traps and sediment barriers, prevent soil loss. Hence this not applicable for project case.
Water consumption and stress	The project operates within natural flow regimes without significant diversion or depletion of local water sources. Optimized reservoir management prevents downstream shortages. Regular assessments and

	stakeholder engagement ensure water availability remains unaffected for local communities and ecosystems. No water stress is experienced in the location due to the project activity. The same has been confirmed during the on-site audit.
Usage of fertilizers	No risk identified.

4.2.8.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Species or habitat	As this project activity is Hydro power plant, this section is not Applicable for this project as no species involved in the project activity.
Areas needed for habitat connectivity	The verification team confirms that the project's design preserves natural water flow, allowing for uninterrupted aquatic migration and wildlife movement. No critical wildlife corridors or breeding grounds have been disturbed, and the surrounding vegetation remains largely unaffected. In addition, preventive measures—such as the establishment of buffer zones and implementation of erosion control strategies—have been effectively applied to support ecological integrity and maintain long-term habitat connectivity.

4.2.8.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	N/A

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	N/A

4.2.8.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion	The project is situated in a non-sensitive area, deliberately avoiding forests, wetlands, and other critical ecosystems. No natural habitats have been cleared, degraded, or altered, as the project design emphasizes environmental sustainability. During On site assessment, the Local

stakeholders confirm that land use remains consistent, with no ecosystem transformation or biodiversity loss observed.

4.3 Accuracy of Reduction and Removal Calculations

The verification team was able to confirm that the monitoring plan contained in registered PD /05/ and MR /1/ is in accordance with the applied methodology for the project activity i.e. AMS I.D “Grid Connected Renewable Electricity Generation” Version 17.0 /06/ and its applicable tools.

The parameter stated in the monitoring plan /05/ and the applied methodology /06/ have been fulfilled in the current monitoring period.

The GHG emission reductions were correctly calculated on the basis of the applied methodology AMS I.D “Grid Connected Renewable Electricity Generation” Version 17.0 /06/ and the formulae given in the monitoring report /1/.

Parameter monitored:

Verification team confirms through On-site audit and from the document review, the actual monitoring system complies with the monitoring plan mentioned in the registered PD /05/. According to the monitoring plan in the registered PD /05/, there is four monitoring parameter required to be monitored.

1. $EG_{\text{facility},y}$ (MWh)

		VVB Assessment
Data / Parameter	$EG_{PJ,y}$	The monitoring parameter, Unit and description are as per the registered PD /05/.
Data unit	MWh/yr	
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh	
Source of data	Electricity meter readings	The monitoring or calculation of this parameter is continuously performed by electricity meters (one main and one check meter) installed at the substation. These meters measure the amount of electricity supplied by the proposed power plant to the national electricity grid with an

		accuracy class/13/ of “0.2s” for both the main and check meters. This was confirmed during the On-site audit, based on the single line diagram/08/, technical specifications of meters /10/, accuracy check /13/ and meter photos /14/. The source of data is in accordance with the registered PD /05/.
Description of measurement methods and procedures to be applied	The difference of final value of export and import is used for monthly values of net electricity supplied to the grid by the project activity and same value will be considered for ER calculations.	The monitoring parameter is continuously performed by electricity meters (one main and one check meter) installed at the substation. These meters measure the amount of electricity supplied by the proposed power plant to the national electricity grid with an accuracy class of “0.2s” for both the main and check meters. This was confirmed during the On-site audit, based on the single line diagram/08/, technical specifications of meters /10/, accuracy check /13/ and meter photos /14/. The source of data is in accordance with the registered PD /05/. Electricity export is recorded monthly /11/ and invoiced /12/ by PP to HPSEB as confirmed during the On-site audit. The measurement methods and procedures are in accordance with the registered PD /05/. The monitoring procedure was verified during the On-site audit and found in accordance with the PD /05/.
Frequency of monitoring/recording	Monthly recording	The electricity export/import is continuously measured and monthly recorded as verified during the On-site audit/physical site visit and from monthly electricity records /11/. The frequency of monitoring/recording is

		in accordance with the registered PD /05/.
Value monitored	72,978	Verification team has verified the reported electricity export/import data with the monthly electricity records /11/ and cross checked from monthly invoices /12/ and found consistent.
Monitoring equipment	The electricity exported / supplied by the plant to pooling substation and further to substation. This meter also measures electricity imported by the plant from the grid. Energy meter with an accuracy: class 1.0 for kWh, class 2.0 for kvarh Energy meter make - DTSD341 Serial Number: 1.Main - 250318494, Backup - 250318495 2.Main - 250318493, Backup - 219014741	Verification team has checked the meter details during the On-site audit/physical site visit, by checking the accuracy check of meters/13/, technical specifications of meters and meter photos /14/ and found consistent with the details mentioned in the monitoring report /1/. The energy meters are in accordance with the registered PD /05/. During the monitoring period, no replacement of meter was observed.
QA/QC procedures to be applied	The frequency of calibration of energy meters is once in 5 years. The monthly electricity supplied/exported by the project activity in the JMR report is cross checked with the monthly invoices of sale. In the absence or delay in the meter calibration appropriate Guidelines will be applied to confirm the conservativeness of metering. The metering arrangement, accuracy class of meters, calibration	Verification team has verified the reported electricity export/import data with the monthly electricity records /11/ and cross checked from monthly invoices /12/ and found consistent. QA/QC procedure has been maintained. As per registered PD /05/, certification of Meters through an accuracy test at least once every year or at any time the readings of Net

	frequency is under control of EDL and PP does not have any control on it.	Energy from the main Meter and Check Meter. There was no delay in the meter accuracy check during this monitoring period as verified from the accuracy check /13/. Further back-up energy meter has been installed for the cross-check purpose in case of any fault in the main meter which is accepted to the verification team.
Purpose of the data	Calculation of baseline emissions	The purpose of the monitoring parameter is to calculate the baseline emission which is in accordance with the registered PD /05/.
Calculation method	Net electricity supplied to the grid by the project plant in a given month = Export(kWh)- Import (kWh)	The calculation method is in compliance with the registered PD /05/.
Comments	Data will be archived in paper & electronic form for two years after the end of crediting period or of the last issuance of VCUERs for this project activity, whichever occurs later.	The project proponent has established a data archiving procedure that ensures all relevant project documentation is maintained in both paper and electronic formats. According to the provided evidence and documented procedures, data will be retained for a minimum of two years following the end of the crediting period or the final issuance of VCUERs, whichever is later. This approach is consistent with the applicable data retention requirements and demonstrates adequate capacity to safeguard the integrity and availability of project records. Therefore, the requirement has been satisfactorily addressed.

		VVB Assessment
Data / Parameter	TEGy	

Data unit	MWh/year	The monitoring parameter, Unit and description are as per the registered PD /05/.
Description	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y	
Source of data	Project Activity Site	The monitoring or calculation of this parameter is continuously performed by electricity meters (one main and one check meter) installed at the substation. These meters measure the amount of electricity supplied by the proposed power plant to the national electricity grid with an accuracy class of “0.2s” for both the main and check meters. This was confirmed during the onsite audit conducted onsite, based on the single line diagram/08/, technical specifications of meters /10/, accuracy check /13/ and meter photos /14/. The source of data is in accordance with the registered PD /05/.
Description of measurement methods and procedures to be applied	Electricity meters	The monitoring parameter is is continuously performed by electricity meters (one main and one check meter) installed at the substation. These meters measure the amount of electricity supplied by the proposed power plant to the national electricity grid with an accuracy class of “0.2s” for both the main and check meters. This was confirmed during the on-site audit, based on the single line diagram/08/, technical specifications of meters /10/, accuracy check /13/ and meter photos /14/. The source of data is in accordance with the registered PD /05/.

		Total electricity produced by the project activity is recorded monthly /11/ and invoiced /12/ as confirmed during the on-site audit. The measurement methods and procedures are in accordance with the registered PD /05/. The monitoring procedure was verified during the on-site audit and found in accordance with the PD /05/.
Frequency of monitoring/recording	Continuous measurement and at least monthly recording	The electricity export/import is continuously measured and monthly recorded as verified during the on-site audit/physical site visit and from monthly electricity records /12/. The frequency of monitoring/recording is in accordance with the registered PD /05/.
Value monitored	72,978	Verification team has verified the reported electricity export/import data with the monthly electricity records /11/ and cross checked from monthly invoices /12/ and found consistent.
Monitoring equipment	The electricity exported / supplied by the plant to pooling substation and further to substation. This meter also measures electricity imported by the plant from the grid. Energy meter with an accuracy: class 1.0 for kWh, class 2.0 for kvarh Energy meter make – DTSD341 Serial Number: 1.Main – 250318494, Backup – 250318495 2.Main – 250318493, Backup – 219014741	Verification team has checked the meter details during the on-site audit/physical site visit, technical specifications of meters /10/, accuracy check /13/ and meter photos /14/ and found consistent with the details mentioned in the monitoring report /1/. The energy meters are in accordance with the registered PD /05/. During the monitoring period, no replacement of meter was observed.

QA/QC procedures to be applied	NA	Not Applicable as per the registered PD
Purpose of the data	To calculate the Project Emissions	The purpose of the monitoring parameter is to calculate the baseline emission which is in accordance with the registered PD /05/.
Calculation method	-	The calculation method is in compliance with the registered PD /05/.
Comments	-	Not Applicable

		VVB Assessment
Data / Parameter	Cap _{PD}	The monitoring parameter, Unit and description are as per the registered PD /05/.
Data unit	W	
Description	Installed capacity of the hydro power plant after the implementation of the project activity	
Source of data	Project Site	The verification team reviewed the nameplate of the project activity and visually confirmed it during the onsite assessment. The installed capacity observed is consistent with that stated in the registered Project Design Document (PD).
Description of measurement methods and procedures to be applied	Determine the installed capacity based on manufacturer's specifications or commissioning data or recognized standards	
Frequency of monitoring/recording	Once at the beginning of each crediting period	
Value monitored	15000000	The verification team has reviewed the commissioning certificate/19/ and project license and confirmed that the reported value is consistent with and aligned to the registered Project Design Document (PD)
Monitoring equipment	Not Applicable	Not Applicable

QA/QC procedures to be applied	Not Applicable	Not Applicable
Purpose of the data	Calculation of project emissions	The purpose of the monitoring parameter is to calculate the project emission which is in accordance with the registered PD /05/.
Calculation method	Not Applicable	Not Applicable
Comments	Not Applicable	Not Applicable

		VVB Assessment
Data / Parameter	A _{PJ}	The monitoring parameter, Unit and description are as per the registered PD /05/.
Data unit	m ²	
Description	Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full	
Source of data	Project Site	The verification team assessed and verified the project activity area using topographical surveys, maps, and satellite imagery, and further confirmed its accuracy through visual inspection during the onsite assessment.
Description of measurement methods and procedures to be applied	Measured from topographical surveys, maps, satellite pictures, etc	
Frequency of monitoring/recording	Once at the beginning of each crediting period	
Value monitored	1520000	Verification team has verified the fuel data and mentioned value has found to be consistent.
Monitoring equipment	Not Applicable	Not Applicable
QA/QC procedures to be applied	Not Applicable	Not Applicable

Purpose of the data	Calculation of project emissions	The purpose of the monitoring parameter is to calculate the project emission which is in accordance with the registered PD /05/.
Calculation method	Directly monitored	The calculation method is in compliance with the registered PD /05/.
Comments	-	-

Opinion: The verification team confirms;

- The monitoring plan has been implemented as per the PD /05/;
- The monitoring complies with the requirement of the applied methodology /06/;
- The information inflow (from data generation, aggregation, to recording, calculation and reporting) is included above under parameter and confirms to the requirement of the PD /05/;
- The values included in the monitoring report and corresponding emission reduction sheet are verified and included under monitoring parameter, wherever appropriate;

Parameters not monitored/ex-ante:

1. Data/Parameter: $EF_{grid,OM,y}$ (tCO₂/ MWh)

Description: Operating Margin CO2 emission factor in year y

	Discussion and verification assessment
Purpose of data	Calculation of Baseline emissions
Verified value	0.4504
Source of value	Calculated as per “Tool to calculate the emission factor for an electricity system, version 07” as 3-year generation weighted average using data for the years 2018, 2019 & 2020
Justification	It is fixed for the entire crediting period as per the registered PD /05/. The value of the Operating Margin in the project activity has been correctly taken as per the “Tool to calculate the emission factor for an electricity system, version 07” and hence accepted by the verification team.

2. Data/Parameter: $EF_{grid,BM,y}$ (Unit: tCO₂/ MWh)

Description: Built Margin CO2 emission factor in year y

	Discussion and verification assessment
Purpose of data	Calculation of baseline emissions
Verified value	0
Source of value	Registered PD /05/
Justification	It is fixed for the entire crediting period as per the registered PD /05/. The value of the Build Margin in the project activity has been correctly taken as per the “Tool to calculate the emission factor for an electricity system, version 07” and hence accepted by the verification team.

3. Data/Parameter: $EF_{grid,CM,y}$ (Unit: tCO_2/MWh)

Description: Combined margin CO₂ emission factor for the project electricity system in year y

	Discussion and verification assessment
Purpose of data	Calculation of baseline emissions
Verified value	0.4504
Source of value	Calculated
Justification	The combined margin emissions factor is calculated as follows: $EF_{grid,CM,y} = EF_{grid,OM,y} * WOM + EF_{grid,BM,y} * WBM$ Where: $EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (tCO_2/MWh) $EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (tCO_2/MWh) WOM = Weighting of operating margin emissions factor (%) = 100% (for simplified CM) WBM= Weighting of build margin emissions factor (%) = 0% (for simplified CM) It is fixed for the entire crediting period as per the registered PD /05/. The value of the ex-ante parameter has been correctly taken as per the registered PD /05/ and hence accepted by the verification team.

4. Data/Parameter: EF_{Res} ($kgCO_2e/MWh$)

Description: Default emission factor for emissions from reservoirs

	Discussion and verification assessment
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Purpose of data	Calculation of Power Density
Verified value	90
Source of value	ACM0002: Grid-connected electricity generation from renewable sources V.21.0
Justification	It is fixed for the entire crediting period as per the registered PD /05/. The value of the Build Margin in the project activity has been correctly taken as per the “ACM0002: Grid-connected electricity generation from renewable sources V.21.0” and hence accepted by the verification team.

5. Data/Parameter: Cap_{BL} (W)

Description: Installed capacity of the hydro power plant before the implementation of the project activity.

	Discussion and verification assessment
Purpose of data	Calculation of Power Density
Verified value	00
Source of value	ACM0002: Grid-connected electricity generation from renewable sources V.21.0
Justification	It is fixed for the entire crediting period as per the registered PD /05/. The value of the Build Margin in the project activity has been correctly taken as per the “ACM0002: Grid-connected electricity generation from renewable sources V.21.0” and hence accepted by the verification team. For new hydro power plants, this value is zero

6. Data/Parameter: A_{BL} (m²)

Description: Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity

	Discussion and verification assessment
Purpose of data	Calculation of project emissions
Verified value	0
Source of value	ACM0002: Grid-connected electricity generation from renewable sources V.21.0

Justification	It is fixed for the entire crediting period as per the registered PD /05/. The value of the Build Margin in the project activity has been correctly taken as per the “ACM0002: Grid-connected electricity generation from renewable sources V.21.0” and hence accepted by the verification team. For new reservoirs, this value is zero
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Opinion:

In the opinion of assessment team, ex-ante parameters value that were applied in the calculations is consistent with PD /05/ and correctly applied in MR /1.2/, emission reductions spreadsheet /2.2/ and justified.

Assessment of Data and Calculation of GHG emission reductions:

According to the approved methodology AMS I.D”, Emission Reductions are calculated as follows:

Baseline Emissions

According to the paragraph 11 of the 17th version of AMS I.D, the baseline emissions are the product of electrical energy baseline $EG_{BL,y}$ expressed in MWh of electricity produced by the renewable generating unit multiplied by the grid emission factor.

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where,

$$BE_y = \text{Baseline emissions in year } y \text{ (t CO}_2\text{/yr)}$$

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

If the project activity is the installation of a greenfield power plant, then:

$$EG_{PJ,y} = EG_{PJ, facility,y} \quad (\text{Equation 2})$$

Where,

$EG_{PJ, facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

Thus, baseline emissions can be calculated as

$$\begin{aligned} BE_y &= EG_{PJ,y} \times EF_{grid,y} \\ &= 72,978 \times 0.45037 = 32,867 \text{ tCO}_2\text{e} \end{aligned}$$

Project Emissions

The project activity is a hydro power project activity and as per the para 39 of the applied methodology the Emissions from water reservoirs of hydro power plants shall be determined considered following the procedure described in the most recent version of “ACM0002: Grid-connected electricity generation from renewable sources”. As per ACM0002: Grid-connected electricity generation from renewable sources V.21.0”, paragraph 35

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} \quad (\text{Equation 1})$$

For this project activity, the emissions from water reservoirs of hydro power plants ($PE_{HP,y}$) are applicable. The power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity (W/m²)

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W) (ie, 15,000,000 W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²) (ie, as per the feasibility report it is 1520000 m²)

A_{BL} = Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero

$$PD = (15,000,000 - 0) / (1,520,000 - 0) = 9.87 \text{ W/m}^2$$

As per ACM0002, version 21.0 methodology, If the power density of the project activity is greater than 4 W/m² and less than or equal to 10 W/m², $PE_{HP,y}$ is calculated

$$PE_{HP,y} = \frac{EF_{Res} \times TEG_y}{1000}$$

Where:

$PE_{HP,y}$ = Project emissions from water reservoirs (t CO_{2e}/yr)

EF_{Res} = Default emission factor for emissions from reservoirs of hydro power plants (kg CO_{2e}/MWh) -As per ACM0002, EF_{Res} table in page number 13, it is 90 kgCO_{2e}/MWh

TEG_y = Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (MWh)

Project Emissions during the Monitoring period:

Vintage period	Net Generation (MWh)	EFRes (kgCO ₂ /MWh)	Project Emissions (tCO ₂ e)
01-September-2023 to 31-December-2023	24,107	90	2,170
01-January-2024 to 31-December-2024	48,871	90	4,398
Total	72,978		6,568

Leakage

As per Para 22 of AMS ID version 17/6/ “If the energy generating equipment is transferred from another activity, leakage is to be considered”. It is denoted by LE_y , however there is no equipment transferred from another activity hence, Hence $LE_y=0$

Emission reductions

Reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y = Emission reductions in year y, t CO₂e

BE_y = Baseline emissions in year y, tCO₂e

PE_y = Project emissions in year y, tCO₂e

LE_y = Leakage emissions in year y, tCO₂e

The results of the emission reduction are shown in the table below:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
01-September-2023 to 31-December-2023	10,857	2,170	0	8,687	0	8,687

01-January-2024 to 31-December-2024	22,010	4,398	0	17,611	0	17,611
Total	32,867	6,568	0	26,298	0	26,298

Opinion:

The verification team confirms that

- The complete data set for the identified and required parameters for the operational days in the current monitoring period was available;
- The reported data has been cross checked with available records, as indicated in the section 4.3 above under monitored data, wherever appropriate;
- The baseline, project and leakage emissions have been determined in accordance with the requirement of the applied methodology, as contained in the final monitoring report and corresponding emission reductions spreadsheet;
- The assumptions, emission factors and default values used are justified, as indicated in the section 4.3 above.

4.4 Quality of Evidence to Determine Reductions and Removals

All relevant documents were checked to assess the correctness and quality of data submitted by the project participants, which are used to determine emission reductions.

All records needed for monitoring are archived in line with the requirements of the registered monitoring plan /05/. No significant lack of evidence and missing data were detected during desk review and On-site audit/physical site visit. Hence, the verification team confirms that the monitoring system ensures required quality of the monitoring system to ensure the quality of the monitored data. All internal data are subjected to QA/QC measures. The monitoring parameter has been measured / determined without material misstatements and is in line with all applicable standards and relevant requirements. The information inflow (from data generation, aggregation, to recording, calculation and reporting) is included in section 4.3 under monitoring parameter and confirms to the requirement of PD /05/.

It was also verified that the plant's team involved in the monitoring of project activity is well experienced. Hence, the verification team concludes that competent staff is employed by the project proponent to carry out the relevant tasks with sufficient accuracy. Furthermore, it was confirmed during On-site audit/physical site visit that training programs /18/ for the monitoring staff are conducted on regular basis.

Evidences (Documents/Interviews) referred for verification of individual monitoring parameter and fixed parameter are defined under section 4.3. We further confirm that, sufficient evidence covering the entire monitoring period and at the required frequency were available. A list of referred documents for verification is also included in Appendix II of this report. Verification team conclude that evidences used to determine the GHG reductions and removals are completely available for the monitoring period (sufficiency of quantity) and are appropriate in quality.

4.5 Non-Permanence Risk Analysis

Not applicable.

5 VERIFICATION OPINION

5.1 Verification Summary

4K Earth Science Pvt. Ltd. (4KES), contracted by “Kosher Climate India Private Limited”, has performed the independent verification of the emission reductions for the VCS project activity (VCS ID- 4588) “Nam Houng – 1 Hydro Power Project” in Sarwari Khad in the state of Himachal Pradesh for the monitoring period 01-September-2023 to 31-December-2024 as reported in the Monitoring Report Version 2.0 dated 14/05/2025. ‘Kosher Climate India Private Limited’ is responsible for the collection of data in accordance with the monitoring plan and the reporting of GHG emissions reductions from the project activity. It is our responsibility to express an independent verification statement on the reported GHG emission reductions from the project activity.

4KES commenced the verification on the basis of the baseline and monitoring methodology AMS I. D “Grid Connected Renewable Electricity Generation” Version 17.0/6/, the monitoring plan contained in the registered PD /05/ and Monitoring Report (Version 2.0 dated 14/05/2025) as per the process described under Section 2 of this report.

4KES verification approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. 4KES planned and performed the verification by obtaining evidence and other information and explanations that 4KES considered necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

In our opinion the GHG emissions reductions reported for the project activity for the period 01-September-2023 to 31-December-2024 are fairly stated in the Monitoring Report Version 2.0 dated 14/05/2025. The GHG emission reductions were calculated correctly on the basis of the

approved baseline and monitoring methodology AMS I. D “Grid Connected Renewable Electricity Generation” Version 17.0 and the VCS standard v4.7 /3/. Verification team confirm that verification of the GHG statement was conducted in accordance with ISO 14064-3: 2019.

5.2 Verification Conclusion

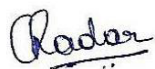
In our opinion the GHG emissions reductions reported for the project activity for the period 01-September-2023 to 31-December-2024 are fairly stated in the Monitoring Report Version 2.0 dated 14/05/2025. The GHG emission reductions were calculated correctly on the basis of the approved baseline and monitoring methodology AMS I. D “Grid Connected Renewable Electricity Generation” Version 17.0 and the VCS standard v4.7 /3/. Verification team confirm that verification of the GHG statement was conducted in accordance with ISO 14064-3: 2019.

Verification period: 01-September-2023 to 31-December-2024(Inclusive of both days)

Verified GHG emission reductions and carbon dioxide removals in the above verification period:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU (tCO ₂ e)	Removal VCU (tCO ₂ e)	Total VCUs (tCO ₂ e)
01-September-2023 to 31-December-2023	10,857	2,170	0	8,687	0	8,687
01-January-2024 to 31-December-2024	22,010	4,398	0	17,611	0	17,611
Total	32,867	6,568	0	26,298	0	26,298

Approved by



Chandrakala R

Managing Director

4K Earth Science Private Limited

Date: 20/05/2025

Place: Bangalore, India

5.3 Ex-ante vs Ex-post ERR Comparison

The comparison of the estimated ex-ante GHG emission reductions and removals and the achieved emission reductions and removals for this monitoring period are reported in the below table.

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
01-September-2023 to 31-December-2023	10,513	8,687	-17%	The seasonal variations in the amount of rainfall and temperature affect the monthly electricity generation from the hydro power plant. Hence, the actual electricity generated from the project is less than the estimated value.
01-January-2024 to 31-December-2024	31,539	17,611	-44%	The seasonal variations in the amount of rainfall and temperature affect the monthly electricity generation from the hydro power plant. Hence, the actual electricity generated from the project is less than the estimated value.
Total	42,052	26,298	-37%	The seasonal variations in the amount of rainfall and temperature affect the monthly electricity generation from the hydro power plant. Hence, the actual electricity generated from the project is less than the estimated value.

APPENDIX I: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification	Assessment method and conclusion
Not applicable	Not applicable	Not applicable	Not applicable

APPENDIX II: LIST OF DOCUMENTS

/1/	Monitoring Report, Version 1.0 dated 06/03/2024 (Initial Version) Monitoring Report, Version 2.0 dated 14/05/2025 (Final Version)
/2/	Draft Emission Reduction Calculation sheet (corresponding to initial Version of VCS MR) Final Emission Reduction Calculation Sheet (corresponding to final Version of VCS MR)
/3/	VCS Standard Version v4.7 https://verra.org/project/vcs-program/rules-and-requirements/
/4/	VCS Program Guide, v4.4 https://verra.org/project/vcs-program/rules-and-requirements/
/5/	Registered PD Web-Link for the Registered project activity: https://registry.verra.org/app/projectDetail/VCS/4588
/6/	Approved monitoring methodology: AMS I D: Grid Connected Renewable Electricity Generation, Version 17.0
/7/	Websites referred: https://www.google.com/maps/@12.9630208,77.7322496,15z?entry=ttu (Latitude-Longitude location finder)
/8/	Single line diagram showing monitoring points
/9/	Organization structure
/10/	Technical Specifications of Power plant

/11/	Monthly electricity exported/imported for this monitoring period.
/12/	Monthly invoices for the electricity exported/imported for this monitoring period.
/13/	Accuracy check for the electricity meters
/14/	Name plate and pictures of meters
/15/	Declaration by PP for No double counting of Emission Reductions
/16/	List of employees
/17/	Comment book, Grievances records for this monitoring period
/18/	Training records
/19/	Commissioning Certificate

APPENDIX III: VERIFICATION FINDINGS

Table 1. Remaining FAR from validation and/or previous verifications

No FAR was raised during previous verification.

FAR ID	XX	Section no.	Date: DD/MM/YYYY
Description of FAR			
Project participant response			Date: DD/MM/YYYY
Documentation provided by project participant			
DOE assessment			Date: DD/MM/YYYY

Table 2. CL from this verification

No CLs.

Table 3. CAR from this verification

CAR ID	01	Section no.	Date: 24/04/2025
Description of CAR			
PP needs to provide or submit the following documents, 1. Double counting declaration. 2. Manufacturing and technical specification of turbines and generators. 3. Calibration of meters and photographs. 4. Meter Invoice 5. Grievance or complaint register photograph. 6. Local Stakeholder consultation.			
Project participant response			Date: 30/04/2025
The following documents have been submitted: 1. Declaration by Project owner on double counting 2. Technical specifications of turbine and generator 3. The frequency of calibration is once in 5 years. Since the project started operations in the year 2023, the calibration of the meters will be done before the year 2028. 4. The invoices for the period September 2023 to December 2024 5. The grievance register maintained at the project site 6. The supporting evidences of local stakeholder consultation conducted on 21/08/2022.			
Documentation provided by project participant			

1. Declaration by project owner 2. Technical specifications of turbine and generator 3. Invoices 4. Grievance register 5. Photographs of LSC meeting	
DOE assessment	Date: 05/05/2025
The submitted documents have been reviewed and found to be accurate and acceptable. CAR 01 is now closed.	

CAR ID	02	Section no.	1.5	Date: 24/04/2025
Description of CAR				
The project representative mentioned in this section does not match the cover page. Please correct it.				
Project participant response				Date: 30/04/2025
<i>The details of the project representative have been made consistent in the cover page and section 1.5 of the MR</i>				
Documentation provided by project participant				
<i>Updated MR</i>				
DOE assessment				Date: 05/05/2025
The project representative's details have been aligned and verified as correct in the revised Monitoring Report. Therefore, CAR 02 is now closed.				

CAR ID	03	Section no.	1.12	Date: 24/04/2025
Description of CAR				
In this section for SDG 8, PP must include the number of jobs created, the total number of employees working, and provide supporting proof.				
Project participant response				Date: 30/04/2025
<i>The number of employment opportunities created with specific to the project activity has been included in the section 1.12 of the monitoring report and the list of employees has been submitted.</i>				
Documentation provided by project participant				
<i>Updated monitoring report</i>				
<i>List of employees</i>				

DOE assessment	Date: 05/05/2025
The number of employment opportunities generated specifically by the project activity is detailed in Section 1.12 of the monitoring report. The submitted list of employees has been reviewed and found to be accurate and acceptable. CAR 03 is closed.	

CAR ID	04	Section no.	2.1.1	Date: 24/04/2025
Description of CAR				
In the "Expected Changes in Well-being" section, PP should also include details related to social well-being.				
Project participant response				Date: 30/04/2025
<i>The social well being has been included in the section 2.1.1 of the monitoring report</i>				
Documentation provided by project participant				
<i>Updated Monitoring report</i>				
DOE assessment				Date: 05/05/2025
Social well-being has been addressed in Section 2.1.1 of the monitoring report and found to be accurate. Therefore, CAR 04 is now closed.				

Table 4. FAR from this verification

No FAR raised during this verification.

FAR ID	xx	Section no.		Date: DD/MM/YYYY
Description of CAR				
NA				
Project participant response				Date: DD/MM/YYYY
Documentation provided by project participant				
DOE assessment				Date: DD/MM/YYYY

APPENDIX IV: TEAM COMPETENCE

<u>Certificate of Competence</u>							
Name	☐ Mr. ☒ Ms.	Swati S Acharya					
Qualification Procedure	Fulfil the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GCC/GHG Projects.						
Appointed to work as:							
	Validator/ Verifier	Team Leader	Trainee	Technical Expert	Technical Reviewer	Financial Expert	Approver
<i>Appointed</i>	Yes	Yes	No	Yes	No	No	No
<i>Appointed Date</i>	30-08-2023						
Authorized to work as Technical Expert for:							
<i>Authorized Technical Area</i>	Sectoral Scope		TA Code		Technical Area within the scope		
	Energy industries (renewable - / non-renewable sources)		1.2		Renewables		
	Transportation		7.1		Transport		
Authorized to work as Local Expert for:							
<i>Country/Countries</i>	India						
<u>Compliance check by:</u>							
				Anand S R			

<u>Certificate of Competence</u>							
Name	☒ Mr. ☐ Ms.	Sriram S					
Qualification Procedure		<i>Fulfil the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GCC/GHG Projects.</i>					
Appointed to work as:							
	Validator/ Verifier	Team Leader	Trainee	Technical Expert	Technical Reviewer	Financial Expert	Approver
<i>Appointed</i>	Yes	No	No	No	No	No	No
<i>Appointed Date</i>	15-06-2024						
Authorized to work as Technical Expert for:							
<i>Authorized Technical Area</i>	Sectoral Scope		TA Code		Technical Area within the scope		
	-		-		-		
Authorized to work as Local Expert for:							
<i>Country/Countries</i>	-						
<u>Compliance check by:</u>							
				Anand S R			

<u>Certificate of Competence</u>							
Name	☐ Mr. ☒ Ms.	Rakshita S Shetty					
Qualification Procedure	Fulfil the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GCC/GHG Projects.						
Appointed to work as:							
	Validator/ Verifier	Team Leader	Trainee	Technical Expert	Technical Reviewer	Financial Expert	Approver
<i>Appointed</i>	No	No	Yes	No	No	No	No
<i>Appointed Date</i>	30-08-2023						
Authorized to work as Technical Expert for:							
<i>Authorized Technical Area</i>	Sectoral Scope		TA Code		Technical Area within the scope		
	-		-		-		
Authorized to work as Local Expert for:							
<i>Country/Countries</i>	-						
<u>Compliance check by:</u>							
Anand S R							

<u>Certificate of Competence</u>							
Name	☒ Mr. ☐ Ms.	Sanjay Kandari					
Qualification Procedure		<i>Fulfil the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GCC/GHG Projects.</i>					
Appointed to work as:							
	Validator/ Verifier	Team Leader	Trainee	Technical Expert	Technical Reviewer	Financial Expert	Approver
<i>Appointed</i>	Yes	Yes	No	Yes	Yes	No	No
<i>Appointed Date</i>	29-08-2023						
Authorized to work as Technical Expert for:							
<i>Authorized Technical Area</i>	Sectoral Scope		TA Code		Technical Area within the scope		
	Energy industries (renewable - / non-renewable sources)		1.1		Thermal energy generation		
	Energy industries (renewable - / non-renewable sources)		1.2		Renewables		
	Energy distribution		2.1		Energy distribution		
	Energy demand		3.1		Energy demand		
	Waste handling and disposal		13.1		Solid waste and wastewater		
	Waste handling and disposal		13.2		Manure		
Authorized to work as Local Expert for:							
<i>Country/Countries</i>	India						
<u>Compliance check by:</u>				Anand S. R.			

APPENDIX V: ABBREVIATIONS

CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
EB	Executive Board
EIA	Environmental Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse Gases
IPCC	Intergovernmental Panel for Climate Change
MP	Monitoring Period
MR	Monitoring Report
MW	Mega Watt
MWh	Mega Watt hour
PD	Project Description
PP	Project proponent
ESPA	Energy Supply and Purchase Agreement
QA/QC	Quality Assurance/Quality Control
tCO ₂	Tonnes of Carbon Dioxide
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCU	Verified Carbon Unit